

Single Variable Calculus Content

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31st October 2025

Functions

- Domain and Range
- Injective, Surjective and Bijective
- Quadratics
- Polynomials
- Trigonometric Functions
- Exponential Functions
- Inverse Functions

Limits and Continuity

- Epsilon-Delta Definition
- One-Sided Limits
- Properties of Limits
- Definition of Continuity
- Properties of Continuous Functions
- Intermediate Value Theorem
- Infinite Limits
- Limits at Infinity

Differentiation

- Definition of Derivatives
- Rules of Differentiation
- Derivatives of Standard Functions
- Derivatives of Inverse Functions
- Chain Rules
- Implicit Differentiation
- Logarithmic Differentiation
- Mean Value Theorem

Applications of Derivatives

- L'Hôpital's Rule
- Monotonicity
- Concavity
- Local Minima and Maxima
- Global Minima and Maxima